

Isometries

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Problem 1. $A : R^n \rightarrow R^n$ preserves distance, ie $d(A(x), A(y)) = d(x, y) := \text{distance from } x \text{ to } y$. Show A takes straight line segments to straight line segments, of the same length (e.g. by generalizing the statement that there is a unique point a distance $d/2$ from two points distance d apart, and it lies on the straight segment between them)

Proof. $\forall x, y \in R^n$, let $d = d(x, y)$. $\forall z \in R^n$ such that $d(x, z) = a$ and $d(z, y) = d - a$ thus z lies on the line segment between x and y . After applying A , $d(A(x), A(y)) = d(x, y) = d$ by definition of A and thus the line segment between $A(x)$ and $A(y)$ are of the same length as x and y . Furthermore, $d(A(x), A(z)) = d(x, z) = a$ and $d(A(z), A(y)) = d(z, y) = d - a$ thus $A(z)$ lies on the line between $A(x)$ and $A(y)$ by the triangle inequality. Thus all points on the line segment xy get mapped to the line segment $A(x)A(y)$ which is the same length. $\square$

Problem 2. Show that A takes triangles to similar triangles. Conclude (by some familiar euclidean geometry) that it preserves angles.

Proof. Let $\forall X, Y, Z \in R^n$. Let $\triangle XYZ$ be the triangle. The line segments of the triangle don't intersect except at X, Y, Z by definition. Since A maps line segments to Line segments of the same length, thus $A(X)A(Y) \cong XY$, $A(X)A(Z) \cong XZ$, $A(Y)A(Z) \cong YZ$ and the lines intersect at X, Y, Z . Thus since the lines are congruent, $\triangle XYZ \cong \triangle A(X)A(Y)A(Z)$ thus A takes triangle to congruent triangles, a stronger condition to similar triangles. Since $\triangle XYZ \cong \triangle A(X)A(Y)A(Z)$, all angles of XYZ are congruent to corresponding angles in triangles $A(X)A(Y)A(Z)$. Thus for any triangle in R^n all of its angles are preserved under A , since all angles are part of a triangle in R^N they are also preserved. $\square$

Now assume $A(0) = 0$. We show A is linear and lies in $O(n)$

Problem 3. Show A preserves norm, ie $\|A(x)\| = \|x\|$, and dot product.

Proof. By definition of A and norm, $\|A(x)\| = d(0, Ax) = d(0, x) = \|x\|$. $\langle x, y \rangle = \|x\|\|y\|\cos(\theta)$. Since all angles and distances are preserved, then $\|x\|\|y\|\cos(\theta) = \|Ax\|\|Ay\|\cos(\theta) = \langle Ax, Ay \rangle$. $\square$

Problem 4. Let B be the matrix with columns $A(e^1), \dots, A(e^n)$. Explain why B is orthogonal. Now consider $Z := B^{-1} \circ A$. Explain why its enough to prove Z is the identity. Explain why Z is rigid, fixes 0 , and all the e^i . Using the above conclude $Z = I$.

Proof.

$$B = \begin{bmatrix} \left| \begin{smallmatrix} A(e^1) \\ \vdots \\ A(e^n) \end{smallmatrix} \right| \end{bmatrix}$$

Since $e_i^T e_j = \delta(i, j)$ since standard basis is orthogonal and since A preserves angles, $(Ae_i)^T (Ae_j) = \delta(i, j)$

$$B^T B = \begin{bmatrix} A(e^1)^T \\ \vdots \\ A(e^n)^T \end{bmatrix} B = \begin{bmatrix} \left| \begin{smallmatrix} A(e^1) \\ \vdots \\ A(e^n) \end{smallmatrix} \right| \end{bmatrix} = \begin{bmatrix} \langle e_1, e_1 \rangle & \cdots & \langle e_1, e_n \rangle \\ \vdots & \ddots & \vdots \\ \langle e_n, e_1 \rangle & \cdots & \langle e_n, e_n \rangle \end{bmatrix} = \begin{bmatrix} \delta(1, 1) & \cdots & \delta(1, n) \\ \vdots & \ddots & \vdots \\ \delta(1, n) & \cdots & \delta(n, n) \end{bmatrix} = I$$

$\forall x, y \in \mathbb{R}^n, \langle Bx, By \rangle = (Bx)^T By = x^T B^T By = x^T y = \langle x, y \rangle$. Thus B is orthogonal.

If we prove that Z is the identity, then Z . Since $O(n)$ is a group, $A, B \in O(n) \implies B^{-1} \in O(n) \implies B^{-1} \circ A \in O(n)$. Since B preserves distance and A preserves distance, B^{-1} preserves distance thus $B^{-1} \circ A$ preserves distance because otherwise A or B^{-1} do not preserve distance and thus causing a contradiction. $A(0) = 0, B(0) = 0$ because B is a matrix thus a linear map, and $B^{-1}(0) = 0$. Because

$$B^T B = I, B^{-1} = B^T. \text{ Furthermore } (B^{-1} \circ A)(e^i) = B^{-1}(A(e^i)) = \begin{bmatrix} \delta(i, 1) \\ \vdots \\ \delta(i, i) \\ \vdots \\ \delta(i, n) \end{bmatrix} = e^i. \text{ Since } Z \text{ fixes origin}$$

and all the standard basis then Z must be I . □

Problem 5. Let $Rig(\mathbb{R}^n)$ be the group of rigid (ie distance preserving) transformations of $\mathbb{R}^n$. Show there is a natural inclusion of $(\mathbb{R}^n, +)$ inside Rig (as translations).

Proof. $\forall v \in (\mathbb{R}^n, +)$, let $f_v : \mathbb{R}^n \rightarrow \mathbb{R}^n \ni \forall w \in \mathbb{R}^n, f_v(w) := w + v$. Since $0, y, x + v, y + v$ form a parallelogram thus $|f_v(y)| = d(f_v(0), f_v(y)) = d(0, y) = |y|$. Thus $f_v \in Rig(\mathbb{R}^n)$ and $g : (\mathbb{R}^n, +) \rightarrow Rig(\mathbb{R}^n)$ s.t $v \rightarrow f_v$. $\forall v, w \in \mathbb{R}^n$ suppose $g(v) = g(w)$ thus $f_v = f_w$ thus $f_v(0) = v = w = f_w(0)$ thus g is injective. □

Problem 6. Let G act on the set X . Show that G^x and G^{gx} are conjugate (recall G^x is the stabilizer of x). Show that $Rig^0 = O(n)$, and that Rig^P is conjugate (in particular, isomorphic) to $O(n)$. Show that Rig is generated by $O(n)$ together with $T := \mathbb{R}^n$ (the translations)

Proof. $G^x = \{f \in G : fx = x\}$, $G^{gx} = \{h \in G : hgx = gx\}$. $\forall h \in G^{gx}, hgx = gx \implies g^{-1}hgx = x \therefore g^{-1}hg \in G^x$. $\therefore g^{-1}G^{gx}g = G^x$. Thus $Rig^0 = O(n)$ and Rig^P are conjugate by the previous statements. Let $f : O(n) \rightarrow Rig^P$ s.t $v \rightarrow (+P)x(-P)$ ($(\pm P)$ is translation which takes 0 to $\pm P$). Conjugation is always a homomorphism. $g : Rig^P \rightarrow O(n) \ni x \rightarrow (-P)x(+P)$. $\forall x \in O(n), g(f(x)) = g((+P)x(-P)) = (-P)(+P)x(-P)(+P) = x$ and $\forall y \in Rig^P, f(g(y)) = (+P)(-P)y(+P)(-P) = y$. Thus f is an isomorphism and $O(n)$ and Rig^P are isomorphic.

By problem 8, $O(n)$ and T generate Rig . □

Problem 7. Which subgroup, T or $O(n)$ in Rig is "better"? Which one is more "canonical"? Which one is normal? Take A , and think of $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Explain why this is differentiable, and prove that the derivative is a constant matrix. Show the map $D : Rig \rightarrow GL_n$ is a group homomorphism, with image $O(n)$. Show the kernel is T .

Proof. T is better because it is normal and is more canonical because it doesn't depend on choice of origin. By problem 8, $\forall A \in Rig(\mathbb{R})$, $A = T_v \circ B$ for $B \in O(n)$ and $T_v \in T$. Thus $\forall T_w \in T$, $A^{-1}T_w A = B^{-1} \circ T_{-v} \circ T_w \circ T_v \circ B = B^{-1} \circ T_w \circ B$. $\forall x \in \mathbb{R}^n$, $(B^{-1} \circ T_w \circ B)x = B^{-1}(Bx + w) = B^{-1}Bx + B^{-1}w = x + B^{-1}w = T_{B^{-1}(w)}(x)$

So $A^{-1}T_w A = T_{B^{-1}(w)}$

A is differentiable, because by problem 8 $\forall A \in Rig(\mathbb{R})$, all transformations are the product of a $B \in O(n)$ with the input vector and then adding a constant vector. The derivative of such a function exists and is B , a constant matrix. $J_A = J_{Bx+v} = B$.

Let $D : Rig \rightarrow GL_n \ni v \rightarrow J_v$. $\forall v, w \in Rig$, $v = T_x \circ B_1$ and $w = T_y \circ B_2$. $D(xy) = x'(y)y'$. However since $x' = B_1$ and $y' = B_2$. Then $D(xy) = B_1 B_2 = D(x)D(y)$. Thus the derivative is a group homomorphism. Moreover $\forall A \in Rig$, $D(A) = B$ but $B \in O(n)$. So $D_*(Rig) \subseteq O(n)$. But $\forall B \in O(n)$, $D(B) = B \in D_*(Rig)$ so $O(n) \subseteq D_*(Rig)$. Thus $D_*(Rig) = O(n)$. $D(A) = 0 \iff A = T_v \circ B = T_v \circ 0$ thus $A = T_v$. Thus $T = \ker D$. $\square$

Problem 8. Show that for any A in Rig , there is an orthogonal matrix B and a vector v , so that $A(X) = BX + v$, ie the composition $T_v \circ B$ (where T_v is translation by v).

Proof. Let A be a rigid transformation. Let $a = A(0)$. Let $B = T_{-a}A$, then $A = T_a \circ B$. $B(0) = T_{-a}(A(0)) = T_{-a}(a) = 0$, thus B fixes the origin and $B \in O(n)$. Expression $A = T_a \circ B$ is unique because since $B(0) = 0$, so $a = A(0)$ and then $B = T_{-a} \circ A$. Thus $A(x) = Bx + a$ $\square$

Problem 9. Compute $A \circ T_v A^{-1}$ $A \in Rig$ (we know T is normal, so this has to be a translation)

Proof. $\forall A \in Rig(\mathbb{R})$, $A = T_v \circ B$ for $B \in O(n)$ and $T_v \in T$. Thus $\forall T_w \in T$, $A^{-1}T_w A = B^{-1} \circ T_{-v} \circ T_w \circ T_v \circ B = B^{-1} \circ T_w \circ B$. $\forall x \in \mathbb{R}^n$, $(B^{-1} \circ T_w \circ B)x = B^{-1}(Bx + w) = B^{-1}Bx + B^{-1}w = x + B^{-1}w = T_{B^{-1}(w)}(x) = T_{D(A^{-1})(w)}(x)$ $\square$

Problem 10. Construct a natural homomorphism from Rig to $+1, -1$ (by taking $\det$ of the derivative).

Proof. Create the homomorphism $f : Rig \rightarrow +1, -1 \ni f = \det \circ D$. Since determinant and D are homomorphisms then f must be a homomorphism. Since $D_*(Rig) = O(n)$ and $\det_*(O(n)) = \pm 1$, $f_*(Rig) = \pm 1$. $\square$